Journal of Environmental Management 277 (2021) 111443



Contents lists available at ScienceDirect

Journal of Environmental Management

journal homepage: http://www.elsevier.com/locate/jenvman



Review



The roles of co-composted biochar (COMBI) in improving soil quality, crop productivity, and toxic metal amelioration☆

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ARTICLE INFO

Keywords:
Biochar
Composting
Soil amendment
Crop production
Toxic metal amelioration

ABSTRACT

The use of co-composted biochar (COMBI) made by the addition of biochar at the beginning of the composting process has greatly increased in agriculture during the last decade. There are more benefits of using the co-composting end product COMBI than using compost and biochar separately or the mixture of the two products. We conducted an extensive review of the production of several COMBIs and their contribution to the composting process and biochar properties as well as the further use of COMBIs in agricultural lands to improve soil health and increase crop yields, and to remediate areas severely contaminated with potentially toxic metals (PTMs). Although the number of researches focused on COMBI production and its application is so far limited, there is enough evidence to elucidate the importance of creating such products to promote sustainable agriculture and environmental safety. Even if a few drawbacks or side effects are found, they are outweighed by the many benefits achieved with COMBIs production and application in comparison to other amendments. The quality of both biochar and compost is largely improved in so many ways during the co-composting process, which in turn improved soil health and crop yields of up to 300% in some particular cases. This work improved the overall understanding of COMBI production and application in agriculture. Based on the review, we suggested future researches to better understand the mechanisms of COMBI long-term application to promote awareness on its role over time through alterations in its surface chemistry, ionic nutrient adsorption, supply (aging effect), and environmental implications.

1. Introduction

In the last two decades, a great effort has been made on researches exploring the benefits of biochar land application to soil and ecosystem health, and the majority of findings has been strongly positive. To a larger extent, the use of compost in agriculture and its many contributions to soil quality in short- and long-term have been well elucidated for many years. Therefore, in the passing decade, some attention has been paid to the application of the mixture of biochar and compost (final products) to soils. However, only a small portion of such researches was dedicated to the addition of biochar to the raw composting materials and its further impact on the composting process, the final co-composted biochar (COMBI) product, and its application to agricultural soils. Most of the findings so far are undoubtedly positive although works focusing on the long-term effects of COMBI application are still scarce and, therefore, more research dedicated to evaluating the potential of COMBI aging in agricultural soils is strongly encouraged.

As a compost additive, biochar improves composting performance, humification process, enhances microbial diversity and activity, reduces greenhouse gas emissions, and immobilizes potentially toxic metals (PTMs) and organic pollutants associated with the compost (Guo et al., 2020). During the composting process, the temperature increases faster in the presence of biochar so does the duration of the thermophilic phase. Since biochar does also enhance the water holding capacity (WHC), the desired moisture content of most composts in the range of 50–60% w/w is ensured when biochar is added at the beginning of the composting process. Different feedstock-derived biochars and composts have different carbon to nitrogen ratios (C/N), which directly affects the speed of organic matter (OM) decomposition (Godlewska et al., 2017). Biochar produced from different methods (pyrolysis or hydrothermal

Abbreviations: COMBI, co-composted biochar; SSA, specific surface area; CEC, cation exchange capacity; WHC, water holding capacity.

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https://doi.org/10.1016/j.jenvman.2020.111443
Received 22 June 2020; Received in revised form 3 September 2020; Accepted 28 September 2020
Available online 10 October 2020
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carbonization) and different feedstocks resulted in various pH profile ranging from basic to acidic. The leachable acidic or alkaline compounds in biochar modifies the pH of its immediate environment, which is more favorable for microbial activity. In addition, large specific surface area (SSA) and high porosity in some type of biochar provide natural protective habitats for microbial proliferation. High cation exchange capacity (CEC) due to certain functional groups in some biochar serves as an electron carrier and facilitates its transferring and transporting (Antonangelo et al., 2019). The enhanced microbial activity to some extent facilitates the mineralization of essential nutrients while reducing gases emission. Not only are the compost itself and its functionality affected by the biochar added in the composting process, but biochar properties are also significantly modified. For example, the CEC of biochars is increased after composting, which is a desirable characteristic for nutrient retention and pH buffering.

As a soil amendment, COMBI has been shown a great performance in improving soil properties and plant growth (Awasthi et al., 2017; Oldfield et al., 2018; Wang et al., 2019), such as alleviating soil drought and salinity stresses (Busch and Glaser, 2015; Lashari et al., 2013, 2015), immobilizing PTMs and organic pollutants (Rodríguez-Vila et al., 2016; Teodoro et al., 2020; Ye et al., 2019), and reducing their bioavailability to soil microorganisms and plants. Such positive results linked to COMBI application are even greater than the effects when simply mixing the final composting product with biochar, probably due to the enhanced reactions of both products during the composting process.

We aim this work to provide recent findings on the composting process as affected by biochar addition, to discuss the final product COMBI, and its application to agricultural and multi-metal contaminated soils. Furthermore, this manuscript is to provide a summary of the most recent studies on using COMBI to improve soil quality and cash crop yields and to immobilize the bioavailability of PTMs to plants and livestock. The main goal is to improve our understanding of COMBI production and application in agriculture and to serve as the basis for future research work. Overwhelming evidences are presented to show the many benefits overtaking the drawbacks of utilizing COMBI to improve soil fertility and plant productivity and yet links the feasibility of applying such product to remediate soils contaminated with potentially toxic metals.

2. Methodology

A comprehensive literature review was conducted of the Web of Science, Elsevier Science Direct, Google Scholar, Scopus, and ProQuest databases using the keywords “biochar, compost, and co-compost” in the title, abstract, and keywords, which identified a total of 362 sources. The search was not limited to certain timespan. The oldest paper found was published in 2008 while the newest one was published in 2020 with the majority published in the last 5 years. From the ProQuest database, which has 26 databases integrated, identical sources were eliminated and it has remained 125 studies out of the 362 found earlier.

The number of publications on co-composted biochars application in soils was also evaluated over time. For this, the search considered three of the most popular databases (Scopus, Science Direct, and ProQuest). Boxplot chart of the number of publications on co-composted biochar application in soils from 2008 to 2019 was plotted and data was verified as per their normal distribution according to the Shapiro-Wilk test ($P \leq 0.05$). The trend line of the number of publications from 2008 to 2019 ($n = 48$) was also verified using the PROC NLIN procedure from SAS ver9.4 to identify the plateau point (joint point) when the number of publications has ceased to increase. The linear regression with the plateau model was only considered when the convergence criterion method from SAS was 100% succeeded. The root mean square error (RMSE) was also calculated to test the model fitness.

3. Production and properties of biochar

3.1. Biochar production

Biochar, a carbon-rich solid material, is formed via thermochemical decomposition of biomass and other carbonaceous materials under limited or no supply of oxygen. Biochar production methods mainly consist of pyrolysis, gasification, torrefaction and hydrothermal carbonization (Enaime et al., 2020). Pyrolysis, either at fast or slow heating rate, results in different product profiles (Pandey et al., 2020). The flash and fast pyrolysis (heating rate of over 200 °C/min for less than 10 s) produces mainly bio-oil with biochar as a by-product. The resultant biochar tends to have low SSA mainly due to entrapment of tar-like material in biochar porous structure. Slow pyrolysis uses longer residence time (around 1 h) and slower heating rate (less than 10 °C/min) which are preferred for high biochar yield and its surface properties due to enough time for re-polymerization of condensable compounds. Microwave assisted pyrolysis is an emerging technique for biochar production due to its high biochar yield (over 60 wt% of the total biomass) and high SSA (between 450 and 800 m^2/g) in produced biochar. In addition, metal and carbon based microwave absorbers can be used to further promote the efficiency of pyrolysis process (Li et al., 2016). Gasification produces mainly gaseous components (syngas or producer gas) containing mainly CO, CO_2, H_2 and N_2 with biochar as a co-product (Qian et al., 2013). However, gasification due to the use of high temperature (commonly from 600 to above 1000 °C) and oxidizing agents (CO_2, air, oxygen, or steam) results in lower biochar yield due to excessive oxidative reactions. Torrefaction is generally performed at slow heating rate and low temperature (200–300 °C), therefore referred as mild pyrolysis or simply thermal pretreatment. Torrefaction removed some volatile compounds such as CO_2, water and oxygen, results in lower energy loss in biomass and higher calorific and heating value in biochar compared to other methods (Zhang et al., 2019). Hydrothermal carbonization (HTC) is conducted by heating biomass in supercritical water at relatively lower temperature (180–250 °C) under high pressure. HTC can process high-moisture biomass without energy intensive drying process in advance (Usman et al., 2019). The biochar produced by HTC process (or hydro-char) has higher char yield (40–70 wt%), higher energy densification ratio (ratio of biochar HHV to biomass HHV), and more oxygen functional groups than pyrolyzed biochar (or pyro-char) (Zhang et al., 2019). Although all these methods are applicable, the most commonly used biochar production method is slow pyrolysis due to its relatively higher yield of biochar with better quality (Kumar et al., 2020).

3.2. Effects of production parameters on biochar yield and properties

The proportion of pyrolysis products (char, bio-oil, gases) depends on the chemical composition of feedstocks and parameters used during production. The pyrolysis process, through release of non-condensable pyro-gases (CO, CO_2, H_2, and lighter hydrocarbons) and oxygenated compounds in bio-oil, reduces the ratio of O/C and H/C in biomass and therefore increases aromaticity and carbon content of the produced biochar (Enaime et al., 2020). Feedstocks with higher content of lignin have higher biochar yield, while biomass with higher content of cellulose and hemicellulose resulted in production of more volatile compounds (Tripathi et al., 2016). Biochar produced from sewage sludge (biosolids) has more hydrogen and less carbon compared to it from lignocellulosic biomass. Due to high amount of inorganic compounds in bio-solids, their derived biochar tends to have more minerals (N and P) and metals (Liu et al., 2018). Pyrolysis parameters such as residence time, feedstock particle size, heating rate, temperature, reactor height, pressure, carrier gas flow rate, and catalyst can affect the properties of the biochar produced (Tripathi et al., 2016). These parameters are interconnected in affecting biochar yield and properties. Longer residence time typically leads to higher yield of biochar due to more chances

for re-polymerization of the biomass constituents. In addition, longer residence time renders biochar more macro- and micro-pores which is a desired trait for land application. Different temperature used in pyrolysis generates biochar that can be used for different applications (Enaime et al., 2020). Pyrolysis with temperature above 500 °C produces biochar with high hydrophobicity, surface area and micro-pore volume, which are suitable to adsorb organic pollutants. At temperature below 500 °C, the produced biochar tends to have more oxygenated-functional groups that are preferable for inorganic pollutants immobilization (Enaime et al., 2020). Using feedstocks with larger particle size can improve biochar yields (Mani et al., 2010), which is due to a larger temperature gradient between the core and outer surface of larger particles. Lower heating rate and temperature reduces depolymerization of biomass into volatile compounds and therefore is preferred for higher biochar yield (Angın, 2013). Increasing height of larger pyrolyzer prolongs vapor residence time and allows re-polymerization, which lead to higher biochar yield. Higher pressure and lower carrier gas flow rate are also reported to enhance biochar yields. Catalysts used in pyrolysis have different effects on biochar yield. Generally, acidic catalysts improve while basic catalysts reduce the biochar yield (Tripathi et al., 2016). Therefore, preliminary examinations should be performed based on feedstock type prior to deciding what production parameters are suitable in biochar formation and developing certain biochar characteristics (porosity, surface area, functional groups, CEC, pH buffering capacity, etc.) (Antonangelo et al., 2019; Sun et al., 2020b). Such characteristics of biochar could significantly affect physicochemical and biological properties of the biochar-amended composting (Fig. 1).

4. Production and characteristics of co-composted biochar (COMBI)

4.1. The effect of biochar on physical properties of compost

4.1.1. Biochar selection for composting

The properties of biochar can affect the efficiency of the composting process. Some critical factors of using biochar in the composting process include biochar loading rate, biochar starting materials and production method, and biochar particle size. The loading rate from 10% to 15% is typically considered optimum, while higher loading rate of 20% (Steiner et al., 2010) and 27% (Chowdhury et al., 2014) had also been reported. Biochar loading rate of high level may result in desiccation in composting due to excessive aeration. The feedstock type and processing temperature for biochar production directly affect properties of biochar, thus affecting the composting process and compost qualities. Woody biomass (e.g. oak wood, beech wood, bamboo) and crop residues (e.g. wheat straw) pyrolyzed at between 400 and 700 °C are mostly reported for production of biochar used in composting (Zhang et al., 2016) because of certain biochar properties favorable for composting such as SSA, porosity, functional groups, CEC, water holding capacity (WHC), etc., are at their optimum levels.

4.1.2. Temperature, moisture, pH during composting

The temperature of composting increased faster when biochar was added than the control without biochar addition (Wei et al., 2014). Biochar addition in some cases caused extension of the thermophilic phase (Chen et al., 2010). Biochar addition also resulted in higher temperature attainment, which was thought to be attributed to less heat loss and higher microbial activities (Li et al., 2015).

Moisture level during composting should be maintained between 50 and 60%. Moisture content at lower level undermines microbial activities while at higher level interferes with air ventilation. Water holding capacity of biochar contributes to maintaining moisture content of composting especially during thermophilic phase (Li et al., 2015). For compost piles with high moisture content, the addition of biochar could absorb excessive water and thus bring the overall moisture content to an appropriate level.

pH usually determines the mobility of heavy metals. The compost with higher pH causes lower solubility of heavy metals, thus reducing its toxicity when used as fertilizer. Biochar in some cases raises compost pH right after its addition due to leaching out of some soluble alkaline compounds in biochar (Li et al., 2015). The pH in some cases decreased more in compost with biochar than without biochar, which was due to enhanced organic and inorganic acids productions from degradation of organic matters (OM) caused by enhanced microbial activities (Wei et al., 2014). The strengthened microbial activities also contribute to a

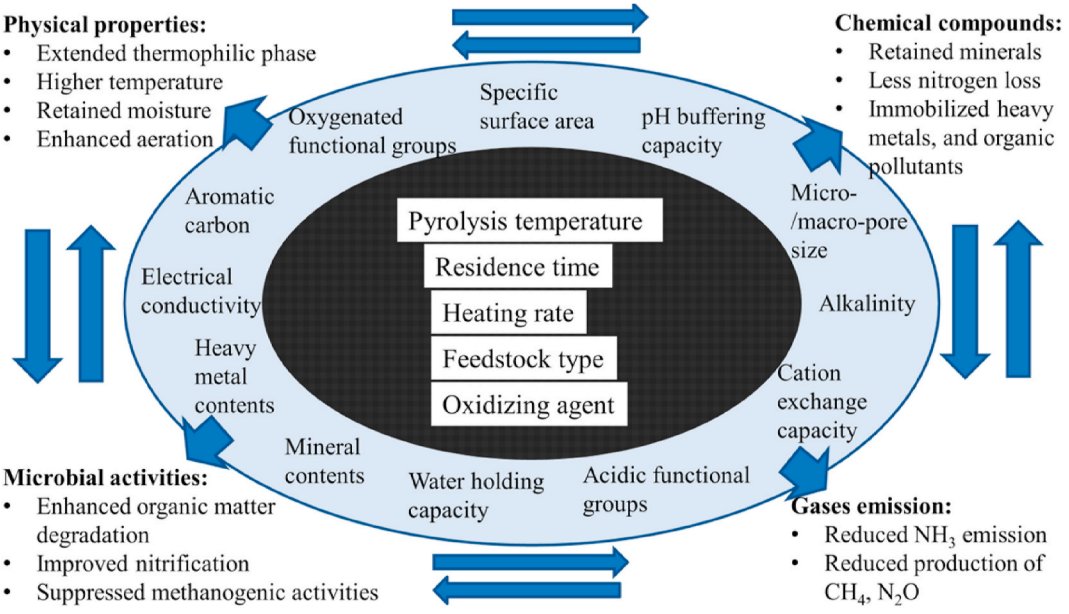


Fig. 1. Schematic diagram of biochar amended composting system. Parameters of biochar production (black solid inner circle) affect biochar physicochemical properties (light blue outer circle) which influence characteristics of the composting process (outside of the circles). The different characteristics of composting also interact with each other (blue two-way arrows) to a certain extent. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

higher degradation of organic nitrogen and production of ammonia at later stage, concomitant with higher pH increase. The produced ammonia/ammonium could be absorbed by negatively charged surfaces on biochar, causing a slight drop in pH. The pH profile in biochar amended composting varies based on properties of biochar, composting method and compositions (C/N, macro- and micronutrients) of composting materials (biosolids, animal waste, yard waste, etc.) (Godlewska et al., 2017).

4.2. Change of organic matter, nutrients, and pollutants in biochar amended compost

4.2.1. Organic matter

Dissolved organic carbon (DOC) was lower in biochar amended compost than it without biochar addition (Zhang et al., 2016). The lower DOC in the compost with biochar was partly due to improved aerobic microbial activities which was caused by enhanced aeration with biochar. In addition, the surface properties of biochar facilitated the adsorption of NH_3, NH_4^+, and H_2S which otherwise could cause inhibitory effects on microbial activities during composting. The chemical absorption of DOC by biochar also reduces effluent discharge from the compost pile. Biochar amended compost has higher stability than the compost without biochar due to the formation of stabilized humic substances (containing fulvic and humic acids) during composting (Jindo et al., 2016). Studies on reduction of organic pollutants during biochar amended composting were fairly limited compared to other pollutants. The most reported organic pollutant is polycyclic aromatic hydrocarbons (PAH) in sewage sludge composting. Biochar showed ability to reduce PAH content in PAH-rich composts (Stefaniuk and Oleszczuk, 2016). The reduction of PAH was thought to be the results of the increased affinity of biochar surface to organic pollutants with biochar aging and improved microbial activities could be the main reason for the reduction of PAH and other organic pollutants (Oleszczuk and Kołtowski, 2018).

4.2.2. Macro plant nutrients

Composting with biochar typically showed lower emission of NH_3 and a higher concentration of NO_3^- compared to that without biochar (Chen et al., 2010; Malińska et al., 2014). With biochar addition, the nitrification process where NH_3/NH_4^+ is converted to NO_3^- by nitrifying bacteria was enhanced, while the ammonification process (the reverse route) was suppressed (López-Cano et al., 2016). In addition, biochar has inorganic phases containing oxides of Fe, Al, Si, phosphates of Ca, Fe, Al, and carbonates of Ca and Mg. These inorganic phases contribute to positively charged biochar surface that is responsible for absorption of NO_3^- (Archanjo et al., 2017). Therefore, biochar amended compost usually shows higher total nitrogen (N) than it without biochar. However, as a second limiting nutrient in soil following N, phosphate (mostly HPO_4^{2-}, $H_2PO_4^-$) content in biochar-amended compost varies among studies (Vandecasteele et al., 2016; Zhang et al., 2016). The improvement of phosphate content in compost with biochar was mostly due to high phosphorus (P) reserves in the biochar added. The reduction of orthophosphate after biochar amendment was mainly attributed to mineralization and microbial utilization of organic P compounds (Vandecasteele et al., 2016). Some nutrients such as potassium (K^+), Ca^{2+}, and Mg^{2+} showed improved retention with biochar amendment was reported due to their adsorption by negatively charged biochar surface. Other elements such as Na^+ didn't show much change after biochar amendment which was considered favorable as Na^+ accumulation can cause deterioration of soil structure (Zhang et al., 2016).

4.2.3. Heavy metals

Biochar has affinity to certain heavy metals such as Zn, Cu, Ni, Pb, As, etc. due to its surface physicochemical properties. The interactions between biochar surface and heavy metals can be classified as physical sorption, complexes formation, precipitation, ion exchange, and electrostatic sorption. Heavy metal profiles in composting material depend on their origin. Sewage sludge and animal waste have shown higher levels of heavy metals than other organic wastes (Liu et al., 2017; Meng et al., 2017; Sun et al., 2018). Composting of sewage sludge and animal waste with biochar showed reduced mobility and bioavailability of heavy metals. It was shown that different temperatures used in biochar production determine biochar's ability for heavy metal immobilization. A study evaluated pyrolysis temperature from 200 to 900 °C, and the biochar produced at temperature between 450 and 500 °C showed better ability in Cu and Zn immobilization during biochar amended composting (Li et al., 2015). However, other studies reported no effects on the bioavailability of heavy metals after composting with biochar (López-Cano et al., 2016) and one reason could be the unfavorable surface properties of the biochar used.

4.3. Change of gaseous emission from biochar amended compost

During composting, several major gaseous compounds such as methane (CH_4), carbon dioxide (CO_2), nitrous oxide (N_2O), and NH_3 can be emitted as the results of degradation of organic matter (Chowdhury et al., 2014). Biochar addition to composting is beneficial in the reduction of CH_4 emission as compared to the case without biochar addition, which is due to the reduced bulk density, better aeration, and gas diffusion, creating growth conditions suitable for methanotrophs (consuming CH_4) while undesirable for methanogens (producing CH_4) (Vandecasteele et al., 2016). Decline and increase of CO_2 emissions after the addition of biochar were also reported (Czekała et al., 2016; Vandecasteele et al., 2016). Declined CO_2 emission was reported attributive to its absorption by biochar, while increased CO_2 emission was due to improved aeration and enhanced degradation of compost substrate. Reduced or less increased emission of N_2O was commonly reported on composting process enhancement with biochar (López-Cano et al., 2016; Vandecasteele et al., 2016). The main reasons are, firstly, biochar absorption of NO_3^- makes the substrate less available for denitrifying bacteria; secondly, metals such as iron and manganese bonded to biochar surface facilitate the chemical reduction of NO_3^- to N_2; thirdly, improved aeration condition facilitates nitrification while suppresses denitrification process and finally, denitrifying bacteria with high expression of N_2O reductase (reducing N_2O to N_2) are enriched during the process. Ammonia is commonly emitted during composting. The emission of ammonia leads to potential green-house gas accumulation and reduction of N in the end compost which undermines its quality as soil fertilizer. Biochar addition in composting to reduce NH_3 emission and prevent N loss has been widely reported (Malińska et al., 2014). Biochar was found to absorb NH_3 and its N precursors (NH_4^+, urea, and uric acid), creating conditions favorable for nitrifying bacteria. It is not uncommon that increased NH_3 emission was reported in some cases when using biochar, which was attributed to the different biochar loading rate (Liu et al., 2017). Due to enhanced aeration, biochar use in composting also reduced emission of H_2S which is a toxic gas from the decomposition of protein and other sulfur-containing organic compounds (Steiner et al., 2010).

4.4. Changes of biochar properties during composting

The composting process also modifies the physicochemical properties of the biochar added to the system (Fig. 1). It was reported that CEC, negatively charged, oxygenated, or acidic functional groups were enhanced in the biochar after composting (Khan et al., 2016; Prost et al., 2013). Enhanced CEC enabled biochar to exchange mineral and metal elements with its environment. Bioavailable K and P were enriched in composted biochar at the cost of reduction of Ca and Mg, as compared to those in non-composted biochar (Prost et al., 2013). Such changes were considered due to concentration gradients of these elements between biochar and its surrounding composting materials. It was also noted, however, that migration of P, K, Ca, and Mg from composting materials

to biochar was much easier than that of heavy metals such as Mn, Cu, Zn, and Mo, regardless of their concentrations (Khan et al., 2016). In this case, heavy metals usually had formed strong complex with organic matter in composting materials. Functional groups such as carboxylic, phenolic, and lactonic groups were increased in biochar with composting time (Hua et al., 2009). These enriched surface groups may be conducive to the immobilization and absorption of nitrogen compounds. Acidic groups such as carboxyl groups could form complexes with NH_4^+. In addition, retention of NO_3^- and DOC on biochar surface limited denitrification process, resulting in the reduction of N_2O and total N loss (Kammann et al., 2015).

Biochar produced at relatively low temperatures from feedstock rich in lignocellulose materials could undergo partial degradation biotically and abiotically (Dias et al., 2010). When such biochar is applied to composting, the increased temperature during the process could facilitate abiotic oxidation of biochar surface, creating more hydrophilic functional groups that are available for microbial degradation (Cheng et al., 2006). Increased pH at later stage of composting was considered to reduce biochar's capacity for retention of phenolic compounds, causing further degradation of biochar materials. The partial degradation of biochar may contribute to the formation of humic acid in the compost pile (Li et al., 2017).

4.5. Microbial consortium in biochar amended compost

Biochar can serve as a protective habitat for microorganisms during composting. The micro- and macro-pores with a size comparable to bacteria provide shelter for them to proliferate with limited stresses from inhibitory compounds (e.g. NH_3, H_2S, and heavy metals), competitors such as pathogens, and environmental factors such as pH, leaching, and desiccation (Sanchez-Monedero et al., 2018). Biochar as a favorable habitat for microbes is due to its retention of DOC, nitrogen compounds and minerals as available microbial nutrients, porous structures for natural barrier and aeration, the water-holding capacity to maintain suitable moisture level, immobilization of toxins to alleviate toxic effects, and pH buffering capacity to damping abrupt pH change during phases of organic acid production and nitrification (Sanchez-Monedero et al., 2018; Sun et al., 2020a, 2020b). Studies have shown the enhanced abundance of bacteria, fungi, and actinomycetes in biochar-added compost compared to compost without biochar (Du et al., 2019).

Microbial community in biochar amended composting has been evaluated by several studies (Du et al., 2019; Wei et al., 2014). In one case, biochar loading rate from 5% to 20% to composting resulted in enriched community of *Bacillus* sp. compared to the system without biochar (Du et al., 2019). In another case, the compost with 1% loading of biochar showed higher abundance of *Flavobacterium* sp., *Actinobacterium* sp., and *Rhizobiales* sp. for degrading lignocellulosic and aromatic compounds, *Acinetobacter* sp. for degrading hydrocarbons, and *Geobacillus thermodenitrificans* for hydrolysis and denitrification (Wei et al., 2014).

The physical, chemical, and biological properties of biochar-amended compost interact with each other in certain way. Biochar properties enhanced aeration and reduced bulk density of its amended compost, thus modified the moisture content, temperature profile, and microbial activities. The enhanced biotic and abiotic activities in turn strengthen degradation of organic matter, heavy metals migration, nitrification and denitrification, formation of surface functional groups and humic substances, further modifying properties of biochar-amended compost (Fig. 1).

5. COMBI application in agricultural lands

5.1. Current conditions of soil in farmlands and new strategies of biochar use

Declining agricultural productivity is severe, especially in the tropics, because of the soil fertility depletion, nutrient imbalances, reduction of soil organic matter (SOM), unsustainable land uses (Agegnehu et al., 2017; Lal, 2015), and the new climate change context leading to a drastic reduction in average global incomes (Aldieri and Vinci, 2020; Kogo et al., 2020), which are directly linked to the growing food insecurity. Furthermore, the study of Bai et al. (2019) supports that climate-adaptive management, which considers both land use and climate change, can help managers make decisions that are more spatially comprehensive at a landscape level and more efficient and sustainable in using resources. The use of biochar provides benefits to land use and to climate change since it contributes to the carbon stock. Therefore, new strategies such as those aiming to enhance the use of biochar as a compost additive to create COMBI should attract more attention.

Soil salinization is also a serious production and environmental issue affecting about 932 M ha of soils worldwide according to Daliakopoulos et al. (2016), which may be alleviated by COMBI addition through the improvement of physical and chemical conditions of salt-stressed soils and reduction of soil salinity by promoting salt leaching (Lashari et al., 2013). Also, to suppress the rapid mineralization of SOM, an alternative to the most popular organic amendments is the use of more stable carbon sources such as carbonized materials or their extracts (Agegnehu et al., 2017). Glaser et al. (2001) showed that carbonized materials from the incomplete combustion of organic materials, such as black carbon, pyrogenic carbon, and charcoal, are responsible for maintaining high levels of SOM and available nutrients in anthropogenic soils of the Brazilian Amazon basin. Yuan et al. (2017) demonstrated that the application of COMBI using on rice hull-derived biochar stabilized soil organic carbon (SOC).

One of the most popular and sustainable soil amendments produced from incomplete combustion is biochar. Its multiple uses, such as the biochar-compost mixture application, are responsible for soil fertility enhancement and crop yields improvement (Agegnehu et al., 2017). Biochar and biochar-compost mixture can improve soil fertility and crop yields by supplying nutrients, increasing soil pH, CEC, nutrient use efficiency (NUE), and nutrient uptake at certain application rate, and also by improving soil WHC especially in sandy soils, and drainage in clayey soils (Agegnehu et al., 2016a; Jeffery et al., 2011). Despite that, studies of the last decade have been shown a more effective way to lead to an accelerated composting process and the production of an end-product with enhanced fertility and carbon sequestration potential, the co-composted biochar (COMBI) as an organic matter source and soil amendment (Fischer and Glaser, 2012; Jia et al., 2016; Sánchez-García et al., 2015; Zhang and Sun, 2014), which will be further explored.

5.2. Co-composted biochar (COMBI) uses in soils

The term co-composted biochar (COMBI) refers to biochar which is mixed with compost feedstock (organic matter that is both rich in nutrients and labile organic carbon, such as sewage sludge, manure, and plant residues) before aerobic composting (Fischer and Glaser, 2012). According to Khan et al. (2016), COMBI may be used to remediate soils contaminated with PTMs and improve soil health since COMBI possesses all the properties biochar and compost have, and more. The same authors have highlighted that the cost of production in co-composting to produce COMBI is even lower than the incubation of biochar + compost mixture, as this latter involves two separate processes, composting and subsequent incubation. Additionally, some negative results from field trials with pure, untreated biochar are somewhat contrary to traditional practices where biochar is used in mixing with manures, human feces,

food waste and agricultural residues (Ogawa and Okimori, 2010) to increase crop yields in poor soils (Kammann et al., 2015).

Recently, the principle of terra preta formation becomes well known as the application of biochar during composting and its utilization as a sustainable soil amendment (Busch and Glaser, 2015; Fischer and Glaser, 2012). In this sense, the fertilizer value of biochars may be enhanced by composting them with organic substrates before their application to soils (Borchard et al., 2014; Prost et al., 2013; Schulz et al., 2013). The co-composted biochar has a great potential to improve crop yields (Bass et al., 2016; Dias et al., 2010; Glaser et al., 2015; Hua et al., 2009; Lashari et al., 2013; Schulz et al., 2013; Schulz and Glaser, 2012; Steiner et al., 2010) although in some cases the impact of COMBI additions on crops yield has been negligible (Schmidt et al., 2014). Recent researches suggested that a minimum proportion of biochar in compost is required to produce a positive effect on plant performance (Schulz et al., 2014).

Overall, the mixing of biochar with any compost feedstock (containing high contents of nutrients and labile organic carbon) before starting an aerobic composting process has been shown to enhance the agronomic performance of biochar as a soil amendment since the produced COMBI slowly releases essential plant nutrients such as N and P (Haider et al., 2016; Kammann et al., 2015). COMBI may be a promising soil amendment material, which not only can overcome the low nutrient content of biochar, but also regulate nutrient release from the compost (Luo et al., 2017; Schulz et al., 2013; Schulz and Glaser, 2012), and reduce leaching of nutrients and contaminants either from inflowing stormwater or from the compost itself (Iqbal et al., 2015).

Co-composting not only reduces the amounts of inhibitory aromatic compounds contained in biochars but also loads them with easily bioavailable C and N sources, which increases their potential fertilizer value. Therefore, like other organic fertilizers, COMBIs might enhance N_2O, CH_4, and CO_2 production after their addition to soils (Borchard et al., 2014). This warrants necessary future researches on the COMBI application to soil and its long-term effect over multiple growing seasons under crop rotation.

The effect of biochar addition on composting processes was extensively investigated by Xiao et al. (2017). Fig. 2 shows the number of publications (NP) from sorted databases on biochar + compost researches since 2007 and elucidates such extensive increase over time until 2020. During this period, most of the attention has been paid to studies in areas of environmental, agricultural, and biological sciences given the importance of their contribution to soil health, environmental quality protection, and crop production mainly in China (15%), United States (13%), Germany (12%), and Australia (11%) (Fig. 3). Meanwhile, fewer studies were focused on the performance of co-composted biochar (COMBI) after land application.

From 2008 to 2019, the ProQuest database had a maximum of 25 publications in one of the years (Fig. 4), which is likely very low in comparison to the maximum NPon either biochar or compost separately in a single year. Upon further analysis, studies on combined compost + biochar and COMBI land-application linearly increased in the last decade until the mid-2016 and then reached a plateau (Fig. 4). This emphasizes that more research must be performed to fill some gaps and to further promote the use of COMBI as a sustainable and efficient agricultural practice worldwide. So far, the majority of studies have found the land application of COMBI takes advantage from both biochar and compost properties to yield better performance in organic and inorganic pollution remediation, NH_3 and greenhouse gas (GHG) emission reduction, soil fertility improvement, and soil biological activity enhancement (Agegnehu et al., 2016b; Wu et al., 2016).

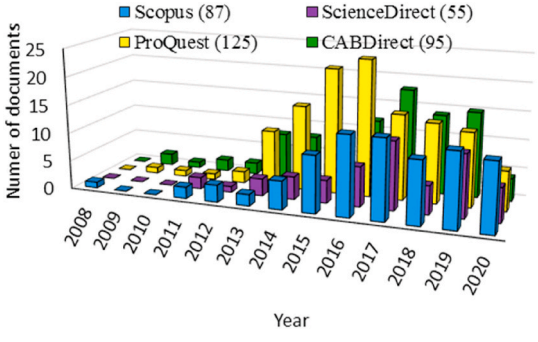


Fig. 2. Number of publications on co-composted biochars application in soils. The search considered three out of the most popular databases (Scopus, Science Direct, and ProQuest). ProQuest has 26 databases integrated. Values in parentheses are the total number of works from 2008 to 2020.

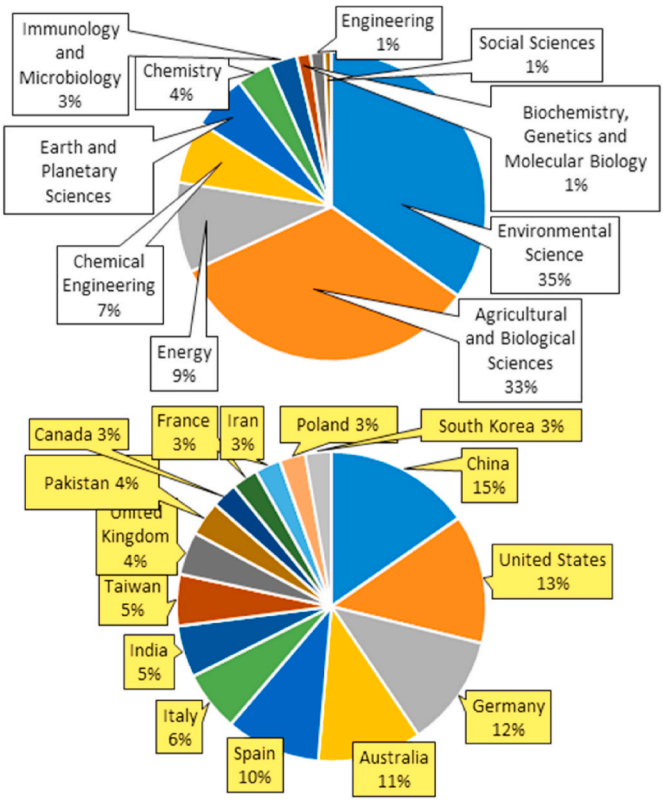


Fig. 3. Distribution of research works on co-composted biochar application in soils by field of research (top) and by country (bottom) from 2008 to 2020. Source: Scopus (2020).

5.3. *Effect of COMBI application on farmland soil properties and crop productivity*

COMBI contains important plant macro- and micro-nutrients depending on the compost and biochar feedstocks, biochar pyrolysis (speed and temperature), and environmental conditions during the composting process. Co-composting biochar with organic waste benefits the compost, the composting process, and the biochar itself by increasing its aging processes. The P, K, and S contents of the COMBI (prepared from chicken manure, pine softwood sawdust, and three biochars from macadamia nutshell, hardwood shavings, and chicken litter) and the CEC of biochars greatly increased in the co-composting process (Khan et al., 2016). Although a considerable amount of research on biochar and biochar-compost mixtures have been conducted, a limited amount of information exists on COMBI applications (Wang et al., 2019). Thus, Table 1 focuses on summarizing the up-to-date impacts of COMBI on soil properties, plant performance, and other benefits associated with COMBI application to farmlands and the most recent findings of benefits to soils and crops after COMBI application with

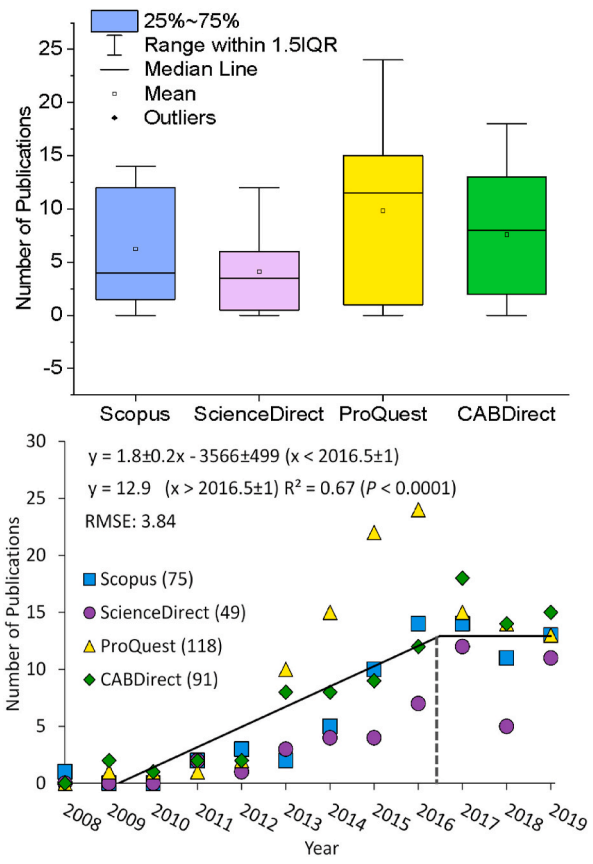


Fig. 4. Boxplot chart of number of publications on co-composted biochar application in soils from 2008 to 2019 (Top). Data are normally distributed as per the Shapiro-Wilk test ($P > 0.05$), except for 'Scopus', ($P < 0.05$). Trend line of number of publications from 2008 to 2019 (Bottom) which successfully converged linear regression with the plateau model (n = 48) (SAS ver9.4, PROC NLIN). The dashed line indicates a joint point (plateau point) when the number of publications ceased to increase. Values in parentheses are the total number of works. RMSE: Root mean square error. The search considered the entire years from 2008 to 2019 to produce both graphs.

biochars derived from different feedstocks. Such findings are discussed next.

Overall, COMBI has been seen to increase C, N, and P in soils and stabilize soil aggregates, as well as stimulate microorganisms (Forján et al., 2018; Mackie et al., 2015; Novak et al., 2019; Oustriere et al., 2017; Radin et al., 2018; Rodríguez-Vila et al., 2016; Schulz et al., 2013; Sizmur et al., 2011; Wu et al., 2016; Ye et al., 2019). The work of Chen et al. (2020) showed that the addition of COMBI not only significantly increased the organic matter and nutrients of a contaminated soil, but also improved its physical and chemical properties, such as electrical conductivity (EC) and pH.

When COMBI was applied to fields at a wide range of application rates in different studies (10–150 Mg ha^{-1}), Wang et al. (2019) predicted using a model that application rates up to 20 Mg ha^{-1} and >30 Mg ha^{-1} significantly increased plant productivity by about 48.3% ($P < 0.001$), but no significant yield increases were found for the application rates between 20 and 30 Mg ha^{-1}. The benefits of applying COMBI might have been ceased out due to increased salt content at those ultra-high application rates (Spokas et al., 2012). According to the authors, such results are promising since it is unlikely that field-scale application of large amounts of COMBI (>20 Mg ha^{-1}) would be economically feasible. Notwithstanding, Schulz et al. (2013) observed that only the application of 100 Mg COMBI ha^{-1} to sandy and loamy soils increased the growth of oat plants. Meanwhile (Schulz and Glaser, 2012), observed the same with the application of up to 90 Mg COMBI ha^{-1}.

When grown with COMBI application, cereal crops (barley, maize, oat, quinoa, and wheat) showed a significant increase in grain yield of 39.7% ($P < 0.001$) in comparison to other treatments not receiving COMBI (Wang et al., 2019). The application of COMBI serves as a partial substitute for chemical fertilizers due to the direct supply of nutrients and the enhancement of nutrients' cycling and availability and stimulation of the microbial population (Sánchez-Monedero et al., 2019). On the other hand, fertile soils may not benefit from COMBI application and a negative response could be due to the dilution effect of the amendment on soil nutrients (Jeffery et al., 2011; Steiner et al., 2010; Wang et al., 2019; Zhang and Sun, 2014). Supportively, promising results have been obtained from the use of COMBI on low fertility soils in Laos (Mekuria et al., 2014).

The COMBI application to a low fertility tropical Ferralsol increased maize grain and biomass production by 10–29% and 9–18%,

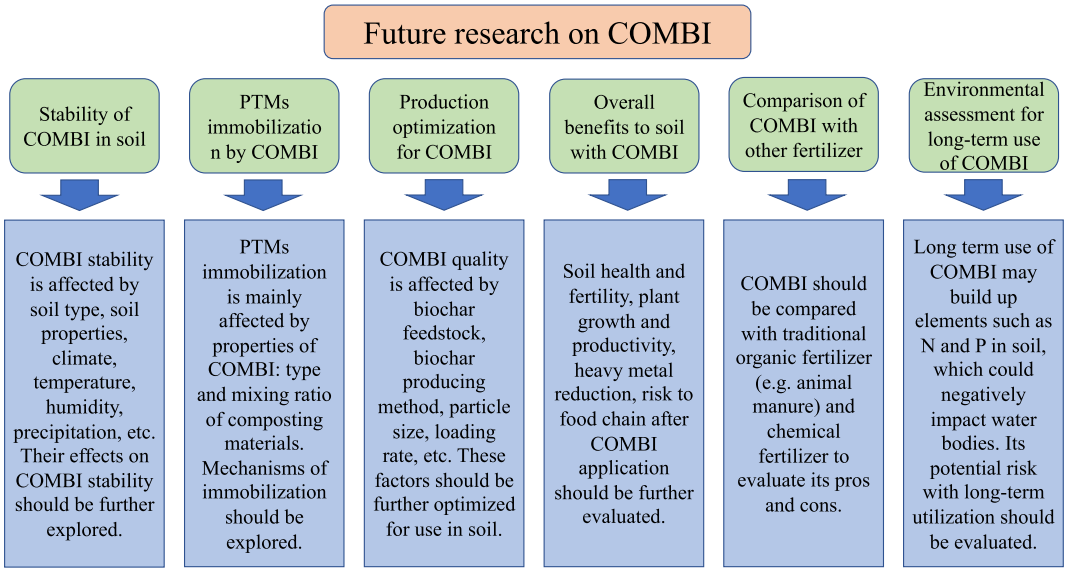


Fig. 5. Recommendations of future researches on production improvement, beneficial use in agricultural lands, remediation of contaminated soils, and long-term environmental benefits and risks of co-composted biochars (COMBI).

Table 1
Effects of co-composted biochars (COMBI) on soil fertility and health, cash crops and other benefits.

Co-composted Biochar (COMBI) feedstock	Effects on soil fertility and soil health	Cash crop	Effects on cash crop	Other benefits	Reference
Waste willow wood (*Salix* spp.)	Increased SOC, available P, Ca and CEC	Corn (*Zea mays*)	Yield increase	Improved soil water retention and nutrient uptake by plants, and lowered N_2O emissions over time	Agegnehu et al. (2016a)
Acacia (*Acacia* spp.)	Increased SOC by 34% and CEC by 24%, and improved soil fertility	Barley (*Hordeum vulgare*)	Yield increase	Improved soil moisture retention	Agegnehu et al. (2016b)
Wheat-straw	Increased the organic nutrients and improved the soil fertility	Chinese cabbage (*Brassica rapa* subsp. *Pekinensis*)	Yield increase	–	Awasthi et al. (2017)
Waste willow wood (*Salix* spp.)	Increased P, Na, K, Mg, NH_4^+, NO_3^-, and soil C stock	Banana (*Musa*) and papaya (*Carica papaya*)	–	Enhanced soil water content and reduced N_2O emissions	Bass et al. (2016)
Mixed woody waste materials	Increased soil available NO_3^- by 100%	–	–	Contributed to soil NO_3^- retention	Hagemann et al. (2017)
Eupatorium adenophorum	Improved soil CEC and K^+, Ca^{2+}, Mg^{2+}	Corn (Zea mays)	Yield increased by 243%	Application of co-composted biochar was economically feasible	Pandit et al. (2020)
Oak residue	Recycled soil C, N, P, and K	Grape (*Viti*), olive (*Olea europaea*), and leek (*Allium ampeloprasum* 'Leek Group')	Yield increase	Resulted in cumulative lower environmental impact	Oldfield et al. (2018)
Garden peat	Increased SOC	Wheat (*Triticum aestivum*)	Increased yield and N and K contents in plants	Reduced CO_2 emission from an alkaline soil	Qayyum et al. (2017)
Oak	Improved soil nutrient contents, pH, and SOC, sustained and enhanced soil biological activity.	Grape (*Viti*) and greenhouse tomato (*Lycopersicon esculentum*)	Enhancement of both crops yield and fruits quality parameters	Partial substitute for mineral fertilizers due to the direct supply of nutrients and nutrients' cycling.	Sánchez-Monedero et al. (2019)
Several (mostly wood-based)	–	Corn (*Zea mayz*), wheat (*Triticum aestivum*), oat (*Avena sativa* L.), and barley (*Hordeum vulgare*)	Yield increased by 39.7%	Plants grown on sandy soils or during a drought benefit from COMBI addition	Wang et al. (2019)

"—": Not given.

respectively, when compared to inorganic fertilizers (Agegnehu et al., 2016a; Pandit et al., 2020). Similarly, in a temperate infertile sandy soil, the addition of 40 Mg ha^{-1} of biochar co-composted with various organic materials showed the same agronomic performance when compared to chemical fertilization (Glaser et al., 2015). The application of highly oxidized COMBI (with high redox potential, Eh) could have a positive agronomic impact on highly reduced soils as well (Pandit et al., 2020). Fig. S1 (Supporting *information*) illustrates the findings of Bass et al. (2016) who applied COMBI to tropical/subtropical soils and further measured its chemical attributes.

Lashari et al. (2015) also demonstrated significant improvement in soil properties, plant performance, and maize yield due to the use of COMBI on a saline soil in China over two years, with potential increased benefits over time. Encouragingly, Lu et al. (2015) found that the application of COMBI combined with pyroligneous acid increased the activities of urease and phosphatase enzymes in saline soils, which alleviated salinity stress on plants. The reason behind the amelioration of saline soils because of the COMBI application was well explored in the works of Lashari et al. (2013) and Lashari et al. (2015). The authors combined COMBI and pyroligneous solution (PS) before their application and verified that both physical and chemical conditions of the salt-stressed soil were improved and thus increased wheat (Lashari et al., 2013) and maize production (Lashari et al., 2015)through a decline in soil salinity. This could be attributed to the multiple benefits on salt leaching, improvement of soil fertility, nutrient supply, and translocation to the grains under the COMBI amendment. Their experiments suggested a great potential to use a combined amendment of poultry manure compost with biochar (COMBI) and the PS from crop straw pyrolysis for alleviating salt stress and sustaining crop productivity in croplands severely threatened by climate change in arid and semi-arid regions globally. However, the mechanisms behind it deserve further research.

Głąb et al. (2018) found that the application of COMBI to sandy soils showed better water retention than the application of compost without biochar. Schulz et al. (2013)showed that COMBI addition to sandy and loamy soil substrates led to increased plant growth with such effect enhanced as the rates of COMBI application increased. The authors attributed those findings to the increased total organic carbon (TOC) content in the soil, as it provides plant-available and mineralizable nutrients. Kammann et al. (2015) observed a biomass yield increase of Chenopodium quinoa of up to 305% in a poor sandy soil amended with 2% (w/w) COMBI. Meanwhile, the growth-promoting effects were more pronounced at lower nutrient-supply levels since COMBI was nutrient-enriched, especially with nitrate (NO_3^-) and orthophosphate. Hence, the authors revealed that when COMBI was applied to soils, the captured NO_3^- in biochar was largely protected against leaching, given the ion-water bonding in micro- and nano-pores of biochar. COMBIs are stable, cost-effective, and can significantly improve soil WHC, aeration, and decrease fertilizer needs and nutrients leaching (Chen et al., 2020).

Busch and Glaser (2015) indicated that COMBI as a soil amendment is an efficient method for carbon sequestration at least in the short-term. According to Khan et al. (2016), composters and farmers are encouraged to produce COMBI and apply it to soils to increase SOC that will persist for a longer period, since the sequestration of recalcitrant biochar carbon is enhanced by producing and applying co-compost to the soil, then mitigating global climate change. This is supported by Steiner et al. (2010), who showed that biochar addition during composting did not change the effective C/N ratio because biochar was largely unaffected by microbial decomposition. Thus, the mineralization rates of labile C remained high, and the biochar contained in the compost provided enhanced durability when applied as a soil amendment with significant implications for C sequestration (Steiner et al., 2010).

Yuan et al. (2017) revealed that the COMBI amendment improved SOC stabilization and reduced CO_2 and N_2O effluxes in addition to improving short-term soil available N, microbial biomass and activity, and the potential of hydrolytic enzymes for degrading organic matter.

Other research has indicated that COMBI did not stimulate N_2O emission; instead, it has led to improved nitrate holding capacity of the soil (Kammann et al., 2015). Darby et al. (2016) further stated that including 1% w/w of biochar to compost before composting process to make COMBI might result in less N_2O emission than the addition of 1% biochar to the final compost product. Likewise, a low COMBI application rate of 1.5% to a costal soil enhanced sesbania and seashore mallow growth and root morphology and increased their biomass by 309 and 70.8%, respectively, while the application rate of 10% w/w significantly inhibited both halophyte growths (Luo et al., 2017). The authors related such findings to the improvement of soil properties including the increased SOM and CEC and the decreased exchangeable sodium (Na) content and the exchangeable Na percentage (ESP).

Considering all the observed effects highlighted above and the acknowledged positive impact of biochar on the composting process, soil application of COMBI represents a promising technical option to increase the sustainability of agricultural systems, which is also stressed by Sánchez-Monedero et al. (2019). In addition, Oldfield et al. (2018) confirmed that COMBI application was environmentally beneficial resulting in a cumulative lower environmental impact when compared to chemical fertilizers. Although some positive effects can be observed by the combined effort of COMBI and chemical fertilizers, Agegnehu et al. (2016a) reported that the COMBI application along with chemical fertilizer augmented the soil macronutrients N, P, and K and increased the peanut yield.

This section outlined the potential benefits of using COMBI as a soil amendment: (i) increasing soil fertility and crop yields, (ii) increasing microbial activity, (iii) reducing soil bulk density, (iv) decreasing N_2O emission, (v) enhancing water holding capacity, (vi) reducing nutrient losses by leaching, (vii) reducing GHG and odor emissions, and (viii) increasing the biochar aging effect. Those benefits were also observed elsewhere in the literature (Akdeniz, 2019; El-Naggar et al., 2019; Godlewska et al., 2017; Teodoro et al., 2020; Wang et al., 2019). Another benefit of using COMBI still barely explored relies on its application as a soil remediation technique to immobilize/reduce PTMs. Such studies deserve some attention and, therefore, will be detailed in the following section.

5.4. Effect of COMBI application on soil and potentially toxic metals (PTMs) amelioration

The synergistic effect of biochar and compost to produce a superior material to be used as a soil ameliorator has been frequently reported (Qayyum et al., 2017; Schmidt et al., 2014; Schulz et al., 2014). According to Wiedner et al. (2015) and Zhang et al. (2014), the surface functional properties of biochars may be increased during composting by enhancing the value of the composting product in the form of fulvic acid and humic acid-like materials. This co-composted material (COMBI) sustains nutrient availability and decreases PTMs phytoavailability in metal-contaminated soils (Agegnehu et al., 2015).

Several authors have proven that the combination of compost and biochar is more suitable than biochar alone to promote the immobilization of PTMs in contaminated soils (Beesley et al., 2014; Chen et al., 2020; Teodoro et al., 2020). As per Teodoro et al. (2020), the integration of biochar into the compost at the production stage results in a maturated soil amendment (COMBI) suitable for soil improvement, with maximized potential as a bio-sorbent for metals. The same author found a decrease in extractable metals (pore water and $CaCl_2$ fractions) from COMBI-amended soil. On the other hand, COMBI addition had no effect on plant development (enzyme activity, chlorophyll production, and uptake of elements) although it positively affected some soil properties.

Different proportions of biochar and compost combinations in COMBI have a significant positive effect on the stabilization of PTMs thus reducing their migration in contaminated soils which otherwise could pose environmental risks (Chen et al., 2020; Luo et al., 2017). Ye et al. (2019) also pointed out that adding biochar to sludge compost could effectively deactivate PTMs and persistent organic pollutants, and inhibit plant migration of PTMs. The latter findings of PTMs remediation and other benefits to soils and crops after COMBI application from different feedstock-derived biochars (Table 2) will be discussed next.

The work of Chen et al. (2020) showed that higher pH was one of the main reasons that COMBI applied treatments had lower bioavailability of PTMs, such as lead (Pb). The authors pointed out that COMBI had a more significant effect on the fixation of Pb than copper (Cu) duo to the fact that DTPA extractable Cu was very high in the sludge compost used. The COMBI amendment also greatly increased the biomass and chlorophyll of Chinese cabbage (*Brassica rapa* subsp. *Pekinensis*) although the treatment amended with chemical fertilizer did not increase significantly (Chen et al., 2020). Finally, they showed that high doses of COMBI containing high N improved the physical and chemical properties of the contaminated soil, reduced the bioavailability of PTMs, and increased plant biomass.

Although the application of COMBI increased Cu sorption and reduced plant Cu content shown by some studies (Borchard et al., 2012; Sizmur et al., 2011), the work of Mackie et al. (2015)observed that it did not immobilize exchangeable soil Cu in a vineyard soil. Additionally, COMBI treatments did not reduce plant Cu content. Some soil exchangeable Cu reduction was found because of significant Cu uptake by cover crops. COMBI increased soil microbial abundance and activity for at least 20 months after the initial application but decreased over time afterwards (Mackie et al., 2015). Therefore, the quality of COMBI and its consequent reduction of pollutant contents and availability of pathogens must be carefully evaluated before large-scale land application (Xiao et al., 2017).

The COMBI application has significantly reduced the concentrations of aluminum (Al), cobalt (Co), Cu, iron (Fe), magnesium (Mg), manganese (Mn), nickel (Ni) and zinc (Zn) in soil pore water in addition to increasing soil pH, total carbon (TC), total nitrogen (TN), and dissolved organic carbon (DOC) content (Rodríguez-Vila et al., 2016). Moreover, the COMBI amendment improved the ability of the mine soils to retain contaminants and enhanced its ability to sustain vegetation, as the *Brassica juncea* (*B. juncea*) did not survive in the untreated soil. The combined use of COMBI and *B. juncea* was an efficient strategy to increase the immobilization of PTMs in mine soils and to reduce the associated contamination risk (Rodríguez-Vila et al., 2016).

Ye et al. (2019), when studying the activated biochar addition to the composting process to make COMBI, demonstrated the efficient performance of COMBI amendment as far as decontamination and detoxification of soil polluted with both PTMs and PAHs in tidal wetlands were concerned. Furthermore, activated biochar with excellent surface characteristics gains significant positive effect during composting which will improve soil structure and nutrient retention, and provide better conditions for microbial growth and activity. COMBI as a biochar-blended co-composting not only decreased the available concentrations of pollutants in contaminated soil but also promoted sustainable agriculture (Ye et al., 2019). Still, there are few studies focusing on the reduction of soil contaminants after the application of compost enhanced with biochar (COMBI), except for the changes in the composting process (Ye et al., 2019). Therefore, the influences of activated biochar addition during the composting process on the remediation efficiency of multi-metal contaminated soils warrant future researches.

In summary, COMBI application is an emerging alternative to the currently available soil remediation techniques to immobilize and detoxify PTMs. The encouragement of further works on COMBI application for soil remediation purposes also relies on the fact that one type of biochar alone may be not appropriate for remediation of multi-contaminated soils (Ahmad et al., 2014; Wu et al., 2017). It has been reported that biochar can effectively reduce Zn and cadmium (Cd) concentrations in soil pore water, but arsenic (As) and Cu concentrations increase due to an increase in pH and DOC (Beesley et al., 2010). In addition, essential plant nutrients such as N, P and Ca may also be immobilized, resulting in nutrient deficiency and further hindering

Table 2
Effects of co-composted biochars (COMBI) on multi-metal contaminated soil, cash crops, and other benefits.

Co-composted Biochar (COMBI) feedstock	Effects on multi-metal contaminated soils and other pollutants	Cash crop	Effects on cash crop	Other benefits	Reference
Mixture of soft wood	Decreased extractable metals (Pb and Zn) from amended soil (pore water and $CaCl_2$ fraction)	Yard grass (*Lolium perenne* L.) and arugula (*Eruca sativa* Mill.)	No effect on plants development (enzymes activity, chlorophyll production, uptake of elements)	Biochar hastened the maturing of woody/green material compost and enhanced soil $CaCl_2$-extractable nutrients	Teodoro et al. (2020)
Wheat-straw	Lowered the bioavailability of Trace metals (TMs), such as Pb	Chinese cabbage (*Brassica rapa* subsp. *Pekinensis*)	Increased the biomass and chlorophyll	Increased pH, OM, N, P, and K of the contaminated soil, and improved EC	Chen et al. (2020)
80% varied hardwood and 20% varied coniferous wood chips	Did not immobilize exchangeable soil Cu content (available fraction)	Cover crops (50% *Lotus corniculatus*, 22% *Medicago lupulina*, 25.4% *Trifolium* spp., 2% *Anthyllis vulneraria*, 0.1% *Hippocrepis comosa*, and 0.5% mix of various herbs)	Did not reduce plant Cu content. Cover crops were taken up significant amounts of Cu within two growing seasons, reducing soil available Cu	Provided no benefits to soil properties, except by increasing extractable organic C. Increased soil microbial abundance and functional activity for 20 months	Mackie et al. (2015)
Holm oak wood	Reduced the concentration of Al, Co, Cu, Fe, Mg, Mn, Ni and Zn in soil pore water, improved the ability of the mine soils to retain contaminants, increased the immobilization of potential toxic elements, and reduced the contamination risk.	*Brassica juncea*	Enhanced the ability of the mine soil to sustain the vegetation when amended	Increased soil pH, total carbon (TC), total nitrogen (TN), and dissolved organic carbon (DOC) content	Rodríguez-Vila et al. (2016)
Corn cob	Decontaminated a soil polluted with both metals and available polycyclic aromatic hydrocarbons (PAHs) in tidal wetlands.	-	–	Positive effect on composting, soil structure, nutrient retention, and microbial growth; ensured sustainable agriculture	Ye et al. (2019)

"—": Not given.

plant growth (Rees et al., 2015; Xu et al., 2016). However, it was reported that the application of COMBI produced by biochar and sewage sludge co-composting increased water-soluble nutrients and reduced bioavailable trace elements (Cu^{2+}, Zn^{2+}, Ni^{2+}, and Pb^{2+}) than the control (Awasthi et al., 2017). Thus, the promotion of COMBI application, which will help reduce PTMs bioavailability and improve soil fertility, becomes necessary and promising since it is inexpensive, sustainable, and exerts synergistic effects on soil remediation and plant growth (Liu et al., 2012; Sizmur et al., 2011; Zhang et al., 2013).

The application of COMBI to soils can immobilize PTMs effectively, and reduce their uptake by crops. However, more attention should be paid to the COMBI application to soils, and the PTMs inactivation mechanisms associated with it. Overall, the investigations on COMBI application remained at the experimental stage, and applications to farmlands in large-scale have not been widely carried out. Therefore, in future studies on COMBI application to soils, both short-term and long-term (>5 years) pot/field experiments are required aiming to evaluate its effects on soil health and COMBI-/soil-derived PTMs and organic pollutants at different time scales. Until this point, few studies have explored and compared COMBI made from biochars derived from different feedstocks and their effects on composting and soil applications. Therefore, more feedstocks for biochar production and corresponding special properties should be evaluated, as ponied out by Guo et al. (2020).

6. Future research needs to use COMBI as an organic fertilizer

(1) The stability of COMBI after applied to soils over a long period of time should be explored to promote its extensive use in the future. Many investigations have confirmed the positive effects of COMBI as a soil amendment. However, different soil types, plant species, and application methods (rates, broadcasted in the soil surface or incorporated, depth of incorporation, and in-furrow) all affect the performances of COMBI in the soil-plant-atmosphere system and the mechanisms behind them should be explored with more details.

(2) The contamination of PTMs may not limit the application of COMBI to soils because the biochar added to the composting materials, if high in PTMs, assists in the PTMs immobilization during composting. The COMBI could also assist immobilizing the PTMs in its surrounding soil that has already been contaminated by heavy metals. The mechanisms behind the immobilization and factors affecting it need to be better understood to apply COMBI more economically and efficiently.

(3) Biochar is one of the key components in COMBI. The feedstock of biochar, biochar production method (gasification, pyrolysis, etc.), and the resulting biochar properties such as particle size, morphological structure, physicochemical properties, and surface functional groups all affect the composting process and the properties of the end composting product. Therefore, a standardized quantification method of those parameters should be established.

(4) The applications of COMBI as organic fertilizer to soil have shown benefits. It remains to be consolidated with more studies if this practice could serve as a multi-use tool, e.g., to enhance the soil health and fertility, improve plant growth and productivity, and reduce the metals bioavailability and their risks to the food chain, when COMBI is applied to effectively detoxify the multi-metal contaminated soils.

(5) More comparison of COMBI with other organic fertilizers such as animal manures and chemical fertilizers as nutrient sources is still necessary. Evaluating the nutrient use efficiency (NUE) of different fertilizers with their increased application rate while growing cash crops of agricultural importance could be performed based on cash crop yields and nutrient uptake.

(6) It should be noted that the consequent increase of soil macronutrients, mainly nitrogen and phosphorus, with the continuous COMBI application might negatively impact water bodies if they

are applied more than crop requirement and accumulated in the soil. Internally, attention must be paid that both biochar and composting materials, depending on their feedstock, might have high amount of macro-nutrients which in their final product (COMBI) can reach significantly higher than the materials used separately. Therefore, optimization of biochar/compost ratio based on their macro-nutrient analysis needs further investigation to achieve its environmental and ecological usefulness (Fig. 5).

7. Conclusions

Application of co-composted biochar (COMBI) to enrich infertile agricultural lands could have enormous benefits both environmentally and economically. The production of COMBI is an effective approach to manage the ever-increasing discharge of organic wastes from human activities includes but not limited to crop and animal production, food processing, and municipal wastewater treatment. Production of biochar as a method of waste management also serve as atmospheric carbon sequestration due to conversion of bio-degradable carbon (biomass) into aromatic carbon (biochar) which is less degradable. Composting as another way of waste management accelerates biological and physiochemical degradation of organic wasting materials in a more organized way. An integrated process by adding biochar in composting would further reduce environmental issues during the waste treatment process by reducing toxic compounds leaching and emission (e.g. H_2S, NH_3, heavy metals) and levels of pathogens, and therefore, making the composting products (COMBI) less toxic and more nutritive for soil fertilization. The produced COMBI with enhanced physicochemical properties is more beneficial in improving soil properties in agricultural lands over biochar and compost applied separately or application of the mixture of the two products. In addition, COMBI when used in arable land has great potential to improve crop productivity and reduce heavy metal contamination. Therefore, the production and use of COMBI to enhance agricultural productivity and protect the environment will be an attractive trend towards a low-carbon agricultural economy in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work is supported by the Arthur L. Reed Endowment Fund at Oklahoma State University.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2020.111443.

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